

MISSOURI



MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 4.2

Null Spaces, Column Spaces,
Row Spaces, and
Linear Transformations

Null Spaces

The null space of an $m \times n$ matrix A is the set of all solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

$\text{Nul } A = \{\mathbf{x}: \mathbf{x} \text{ is in } \mathbb{R}^n \text{ and } A\mathbf{x} = \mathbf{0}\}$

Equivalently, $\text{Nul } A$ is the set of all $\mathbf{x}$ in $\mathbb{R}^n$ that are mapped into the zero vector of $\mathbb{R}^m$ via the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$.

Example 1

Show that $\text{Nul } A$ is a subspace of $\mathbb{R}^n$.

Theorem

The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.

Equivalently, the set of all solutions to a system $A\mathbf{x} = \mathbf{0}$ of m homogenous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

Example 2

Determine if $\mathbf{w} = \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix}$ is in $\text{Nul } A$, where

$$A = \begin{bmatrix} 5 & 21 & 19 \\ 13 & 23 & 2 \\ 8 & 14 & 1 \end{bmatrix}$$

Example 3

Determine whether

$$W = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} : \begin{array}{l} a + 3b = c \\ b + c + a = d \end{array} \right\}$$

is a vector space.

Explicit vs. Implicit Descriptions of Nul A

When a vector space V is described as the null space of a matrix A , we have an implicit description of V since V is defined by a condition that must be checked.

To find an explicit description of Nul A , we simply solve the system to find a set of vectors which span the space.

Example 4

Find an explicit description of Nul A by listing vectors which span the null space.

$$A = \begin{bmatrix} 1 & 5 & -4 & -3 & 1 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Comments About Null Spaces

The spanning set for Nul A produced by our standard solution technique is automatically linearly independent.

When Nul A contains nonzero vectors, the number of vectors in the spanning set for Nul A equals the number of free variables in the equation $A\mathbf{x} = \mathbf{0}$.

Column Spaces

The column space of an $m \times n$ matrix A is the set of all linear combinations of the columns of A .

If $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$, then
 $\text{Col } A = \text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$

$\text{Col } A$ is a subspace of $\mathbb{R}^m$.

Example 5

Show that W is a subspace of $\mathbb{R}^4$ by showing that $W = \text{Col } A$ for some matrix A .

$$W = \left\{ \begin{bmatrix} b - c \\ 2b + c + d \\ 5c - 4d \\ d \end{bmatrix} : b, c, d \text{ real} \right\}$$

Comments About Column Spaces

The column space of an $m \times n$ matrix A is all of $\mathbb{R}^m$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution for each $\mathbf{b}$ in $\mathbb{R}^m$.

$\text{Col } A$ is the range of the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$.

The Row Space of a Matrix

The row space of an $m \times n$ matrix A is the set of all linear combinations of the rows of A , where each row of A can be viewed as a vector in $\mathbb{R}^n$.

Note that $\text{Row } A = \text{Col } A^T$.

We generally write row vectors in $\mathbb{R}^n$ horizontally as ordered n -tuples.

Example 6

Find a nonzero vector in $\text{Nul } A$, a nonzero vector in $\text{Col } A$, and a nonzero vector in $\text{Row } A$.

$$A = \begin{bmatrix} 4 & 5 & -2 & 6 & 0 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix}$$

Contrasting $\text{Nul } A$ and $\text{Col } A$ where A is $m \times n$

Nul A

- Is a subspace of $\mathbb{R}^n$
- Implicitly defined
- It takes time to find vectors in $\text{Nul } A$.
- There is no obvious relationship between entries in A and $\text{Nul } A$.
- A typical vector $\mathbf{v}$ in $\text{Nul } A$ has the property that $A\mathbf{v} = \mathbf{0}$.

Col A

- Is a subspace of $\mathbb{R}^m$
- Explicitly defined
- It is very easy to find vectors in $\text{Col } A$.
- There is a very obvious relationship between entries in A and $\text{Col } A$.
- A typical vector $\mathbf{v}$ in $\text{Col } A$ has the property that $A\mathbf{x} = \mathbf{v}$ is consistent.

Contrasting Nul A and Col A where A is $m \times n$

Nul A

- Given a specific vector $\mathbf{v}$, it is easy to tell if $\mathbf{v}$ is in Nul A . Just compute $A\mathbf{v}$.
- Nul $A = \{\mathbf{0}\}$ iff $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- Nul $A = \{\mathbf{0}\}$ iff the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.

Col A

- Given a specific vector $\mathbf{v}$, it may take effort to tell if $\mathbf{v}$ is in Col A since row reduction of $[A \ \mathbf{v}]$ is needed.
- Col $A = \mathbb{R}^m$ iff $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b} \in \mathbb{R}^m$.
- Col $A = \mathbb{R}^m$ iff the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^m$.

Linear Transformations

A linear transformation T from a vector space V into a vector space W is a rule that assigns to each vector $\mathbf{x}$ in V a unique vector $T(\mathbf{x})$ in W such that

- $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in V , and
- $T(c\mathbf{u}) = cT(\mathbf{u})$ for all $\mathbf{u}$ in V and all scalars c

Linear Transformations

If $T: V \rightarrow W$ is a linear transformation, then

- The kernel of T is the set of all $\mathbf{u}$ in V such that $T(\mathbf{u}) = \mathbf{0}$, and
- The range of T is the set of all vectors in W of the form $T(\mathbf{x})$ for some $\mathbf{x}$ in V .

If T is a matrix transformation $T(\mathbf{x}) = A\mathbf{x}$, then the kernel of T is the null space of A and the range of T is the column space of A .

Example 7

Define $T: \mathbb{P}_2 \rightarrow \mathbb{R}^2$ by $T(\mathbf{p}) = \begin{bmatrix} \mathbf{p}(0) \\ \mathbf{p}(1) \end{bmatrix}$.

- a) If $\mathbf{p}(t) = 3 + 5t + 7t^2$, find $T(\mathbf{p})$.
- b) Show that T is linear.
- c) Find a polynomial $\mathbf{p}$ in $\mathbb{P}_2$ that spans the kernel of T and describe the range of T .
